Spanish Media Monitor

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1 Project Summary

This project develops a system for monitoring bias and informational pluralism in Spanish television newscasts using text analysis techniques and language models. The idea is inspired by international initiatives on media bias detection.[1] The main goal of the associated web platform is to provide visualizations of bias and pluralism trends for the general public.[2]

2 Data and Content Classification

2.1 Data sources

TV channels. I collect daily the midday and evening newscasts of four national high-audience channels: *Televisión Española* (TVE, public), *Antena 3* (A3), *La Sexta* (La6), and *Telecinco* (T5). Coverage starts in December 2022 and is updated continuously. Videos are converted to audio and transcribed using *speech-to-text*. These newscasts are comparable in format, duration, and schedule, which facilitates high-frequency analysis.

Agencia EFE (exogenous news supply). As a *proxy* for the daily mix of available stories, I download all Spanish-language pieces from the EFE agency (32,525 during the period covered by the manuscript). For each day d, I record the set of stories and their political composition; these measures serve (i) as supply instruments and (ii) as a benchmark to quantify the direction in which TV channels deviate in their editorial treatment.

[1] For example, the *Media Bias Detector* from the University of Pennsylvania.

[2] See the public demo of the monitor: https://luisignaciomenendez.github.io/media_monitor/index.html.

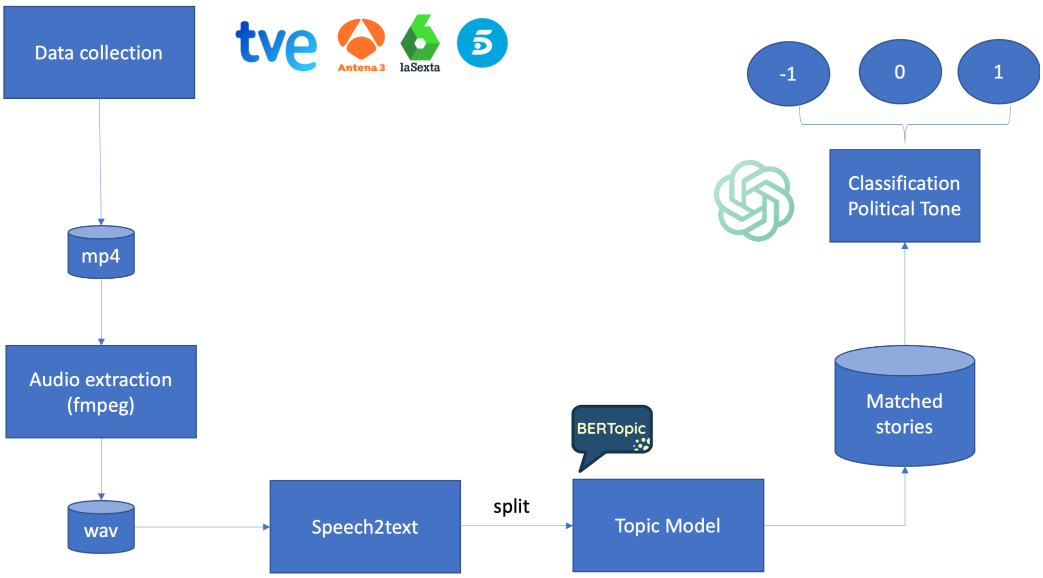
2.2 Preprocessing and segmentation

From the transcripts, I divide each newscast into segments ("stories") using `BERTopic` applied to text and temporal metadata, so that each story s captures a coherent topic. This story-level segmentation provides sufficient semantic context for classification. In total, more than 20,000 stories are obtained; the political subsample is defined using a dictionary of terms (parties, leaders, and general vocabulary).

Pipeline description

This appendix summarizes the end-to-end *pipeline* (Figure 1): storage of daily broadcasts, audio extraction and transcription; segmentation into one-minute stories; topic estimation with a topic model; and political tone classification with an LLM. The corpus includes midday and evening newscasts from TVE, A3, La Sexta, and T5, totaling 563 hours, stored and versioned in Google Cloud Storage (GCS).[3]

Figure 1: Pipeline for content download and classification



Note: Videos are downloaded daily, audio is transcribed with Google, segmented by minute, and stories are grouped with `BERTopic`. Finally, an LLM (GPT-4) classifies the political tone.

Audio and transcription. Each broadcast is converted from MP4 to mono PCM WAV to standardize the input for automatic speech recognition. Transcriptions are obtained with

[3]Project supported by Google Cloud for Education credits and OpenAI research credits.

Google Cloud Speech-to-Text (language `es-ES`, *long-running*, automatic punctuation). The service returns time stamps; I store both the raw JSON and a normalized text version.

Segmentation and alignment. Newscasts are divided into one-minute stories using program markers and textual signals (presenter transitions and on-screen captions).

Topic discovery. Topics are estimated with *BERTopic* and multilingual *sentence embeddings* (`paraphrase-multilingual-MiniLM-L12-v2`). After fitting, I apply outlier reduction and topic refinement; residual or incoherent clusters are excluded from matched comparisons.

Ideological classification with LLM. The political tone of each story is obtained with an LLM on a discrete grid by party. To reduce costs, I first filter using a keyword dictionary; the final set contains 15,406 stories. The *prompt* asks to rate tone with respect to PP, PSOE, Vox, and Podemos/Sumar using values in $\{-1, -0.5, 0, 0.5, 1\}$, where -1 is very negative, 0 neutral/irrelevant, and 1 very positive. Scores are aggregated at the program–day level, weighted by duration (seconds per story). Channel summaries average across program–days, and matched comparisons are restricted to aligned topic–day pairs.

3 LLM Classification and Variable Construction

Each political story $s \in P_{jd}$ (channel j, day d) is classified with an LLM as:

$$t(s) \in \{-1, 0, 1\},$$

where 1 denotes positive tone and -1 negative tone. If $t(s) \in \{-1, 1\}$, a block $p(s) \in \{L, R\}$ (left/right) is also assigned. The classification captures thematic *framing* without requiring explicit mentions of parties or leaders.

Let $\ell(s)$ be the duration of story s (in minutes) and L_{jd} the total duration of the newscast. I define the content characteristics at the program–day level as *time shares*:

$$x_{jd}^{\text{party}+} = \frac{1}{L_{jd}} \sum_{s \in P_{jd}} \mathbf{1}\{t(s) = 1\}\, \mathbf{1}\{p(s) = \text{party}\}\, \ell(s), \tag{1}$$

$$x_{jd}^{\text{party}-} = \frac{1}{L_{jd}} \sum_{s \in P_{jd}} \mathbf{1}\{t(s) = -1\}\, \mathbf{1}\{p(s) = \text{party}\}\, \ell(s), \tag{2}$$

$$x_{jd}^{\text{pol}} = \frac{1}{L_{jd}} \sum_{s \in P_{jd}} \ell(s), \tag{3}$$

for party $\in \{L, R\}$. These variables capture *intensity* (minutes) and *tone* (sign) and are used as product characteristics in the demand estimation.

Supply instruments using EFE. I replicate the same pipeline on all EFE stories for day d to construct the exogenous political mix:

$$z_d^{\text{party}+} = \frac{1}{|N_d|} \sum_{s \in P_d} \mathbf{1}\{t(s) = 1\} \, \mathbf{1}\{p(s) = \text{party}\}, \tag{4}$$

$$z_d^{\text{party}-} = \frac{1}{|N_d|} \sum_{s \in P_d} \mathbf{1}\{t(s) = -1\} \, \mathbf{1}\{p(s) = \text{party}\}, \tag{5}$$

$$z_d^{\text{pol}} = \frac{|P_d|}{|N_d|}, \tag{6}$$

where N_d is the total number of EFE stories and P_d the political subsample. These z_d summarize the daily exogenous supply and serve to instrument x_{jd} in the demand estimation.

4 Informational Diversity and Entropy

A central concept in the web monitor is **entropy**, which measures how diverse and balanced the information environment is. Intuitively, entropy captures the degree of unpredictability in media coverage: a high-entropy day means that news outlets covered a wide variety of topics and perspectives, whereas low entropy indicates concentration on a few narratives or viewpoints.

Formally, if p_i denotes the share of time devoted to topic or political block i, informational entropy is defined as:

$$H = -\sum_i p_i \log(p_i).$$

Entropy is maximal when all perspectives receive similar attention and minimal when coverage is dominated by one side. In the context of media pluralism, higher entropy signals greater informational diversity and lower risk of editorial polarization.

On the platform `mediamonitor.es`, entropy is calculated daily for each channel and visualized alongside tone and bias indicators. Users can thus observe how content diversity evolves over time, especially around key political events or campaign periods.

5 Web Structure and Methodological Tabs

The web interface is designed as an interactive summary of the analytical pipeline described above. Each tab corresponds to a different stage or metric of the monitoring system:

- **Home (Overview)**: introductory visualization showing recent political balance and total volume of news analyzed.
- **Word Counts (Wordcounts)**: displays the most frequent words and entities mentioned across channels and days. Viewers can compare linguistic focus and framing intensity.
- **Topic Model (Topics)**: groups stories using BERTopic, enabling exploration of thematic clusters (e.g., economy, environment, international politics).
- **Tone Classification (Tone)**: shows results from the LLM-based political tone classifier by party and day, distinguishing positive and negative coverage.
- **Bias and Slant (Bias)**: presents the estimated ideological slant index, summarizing the relative tone balance toward left and right parties as defined in equation (7) of the manuscript.
- **Entropy (Entropy)**: visualizes the evolution of informational diversity across channels and periods, highlighting moments of convergence or polarization in coverage.

Together, these components provide a transparent view of how Spanish television newscasts allocate attention and tone over time, combining word-level metrics, topic modeling, and information-theoretic measures to track pluralism and bias in real time.

5.1 Bias Index (*slant*) and Validation

I define the *slant* index of channel j as the net tone balance by blocks:

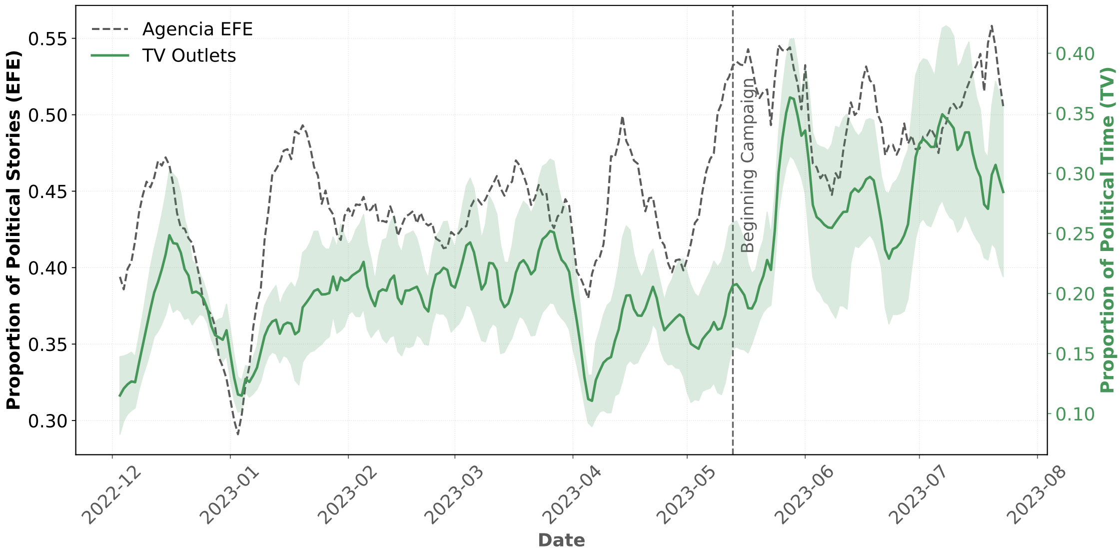
$$\text{Slant}_j \equiv \left(\bar{x}_j^{R+} - \bar{x}_j^{R-}\right) - \left(\bar{x}_j^{L+} - \bar{x}_j^{L-}\right), \tag{7}$$

where bars denote time averages. Positive (negative) values indicate relative right-leaning (left-leaning) bias.

Political Coverage and Electoral Campaign

Below I show descriptive statistics to illustrate the type of analysis enabled by the project, focusing on the 2023 campaign. Campaigns concentrate public attention on politics and shift newsroom priorities. Consistent with this, Figure 2 shows that the share of political content rises sharply in both television and EFE: on average, TV channels increase from 18% outside the campaign to 30% during it; in EFE, the proportion of political stories rises from 43% to 51%. Both series exhibit similar seasonality (Pearson correlation $r = 0.59$), with drops in non-political periods such as Christmas and Easter.[4]

Figure 2: Share of Political Coverage Over Time

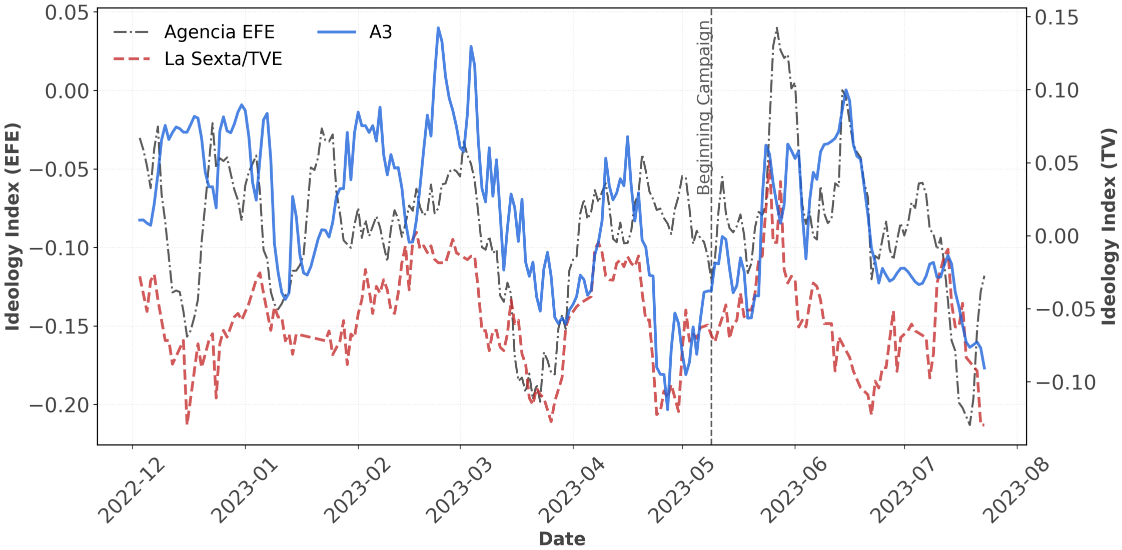


Note: Share of political content for TV (dashed line) and EFE (solid line). Series smoothed with a centered 9-day moving average.

Clear structural breaks are visible. In March 2023, A3 (right-leaning) experiences a reversal in its *slant* index, temporarily converging toward the left-leaning channels. This shift coincides with an increase in negative stories about the right and positive ones about the left (Vox's failed no-confidence motion, internal divisions in the opposition, debates on gender identity, euthanasia, and equality laws; accusations of blocking judicial reform; and, in parallel, favorable government narratives on employment, housing law, and industrial announcements). A3 does not recover its previous bias until June, after the right's strong results in the May regional elections, when divergence and polarization between blocs re-emerge.

[4]There is heterogeneity in agency alignment: A3 co-moves more with EFE ($r = 0.67$), followed by T5 ($r = 0.48$), TVE ($r = 0.38$), and La Sexta ($r = 0.26$). The channel-level breakdown is provided in the Appendix.

Figure 3: Evolution of the *Slant* Index



Note: Time series of the *slant* index (see (7)) for TV (right axis) and EFE (left axis). Positive values: pro-right coverage; negative: pro-left. Channels are grouped into left (La Sexta, TVE) and right (A3). Series smoothed with a centered 9-day moving average.